

Title: Soil Carbon Sequestration Under Regenerative Agriculture Practices

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Abstract

Regenerative agriculture practices, including reduced tillage, cover cropping, and diverse crop rotations, are promoted as strategies for increasing soil organic carbon. This study measures soil carbon accumulation over an eight-year period across matched conventional and regenerative farms in temperate cropland systems.

1. Introduction

Soil carbon sequestration is frequently cited as a scalable climate mitigation strategy, yet field-based estimates of sequestration rates under real-world farming conditions remain limited. This study addresses that gap through a paired-farm longitudinal design.

2. Methods

Soil samples were collected annually from eighteen paired farm sites, each pair consisting of one farm using regenerative practices and one using conventional tillage and monoculture rotation. Soil organic carbon was measured at three depth intervals down to 60 centimeters.

3. Results

Regenerative farms accumulated soil organic carbon at an average rate of 0.42 tonnes per hectare per year in the top 30 centimeters, compared to a slight net loss on conventional farms over the same period. Gains were concentrated in the first four years and showed signs of plateauing thereafter.

4. Discussion

The observed plateauing effect suggests that soil carbon sequestration under regenerative practices may have a finite capacity within a given management regime, with implications for how these practices are credited in carbon offset schemes.

5. Conclusion

Regenerative practices produce measurable soil carbon gains, but claims of indefinite linear sequestration are not supported by this dataset. Policy frameworks should account for diminishing returns over time.